

When maximality derives kinds and contrastiveness: the case of Korean *-(n)un*

1. Main puzzle

- A so-called topic marker in Korean, *-(n)un*, is one of the most extensively researched phenomena in Korean linguistics.
- Most recent work on *-(n)un* has focused on identifying the semantics of contrastiveness which it gives rise to (Lee 2003, Kim 2018a,b).

(1) A: Do you have money?

B: *na tongceon-un iss-e.*

I coin-TOP have-DECL

‘I have COINS[†].’

↷ e.g. I don’t have bills.

(Lee 2003)

[†] Uppercase letters indicate prosodic prominence, which in English gives rise to contrastiveness in some contexts.

- The other use of *-(n)un*, which has not yet been clearly studied, is at best described as the marking of *-(n)un* that does not give rise to such contrastive implication, thus often referred to as ‘neutral’ or ‘non-contrastive’ *-(n)un*.

(2) *say-nun nal-n-ta.*

bird-TOP fly-PRS-DECL

‘Birds fly.’

(Lee 1996)

- Previous research shows two limitations: (i) they do not provide clear definition of the non-contrastive use; (ii) more importantly, they do not account for the fact that *-(n)un* is used in two seemingly very distinct ways.

In this paper, ...

- I. I make an empirical observation that *-(n)un* shows sensitivity to the type of NPs that it appears with.
- II. I propose a single unified function of *-(n)un*, namely marking maximality of the NP it attaches to, which derives both contrastive and non-contrastive uses of *-(n)un* while interacting with the different interpretation of the NP.

- Outline:

§2. A quick note on data

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§3. Two observations

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2. A quick note on data

- Korean is disglossic in that it shows different strategies in how the arguments are morphologically marked (e.g. *-(n)un*-marking), depending on whether it is in a written or a spoken form.
- In written Korean every argument is required to be marked with some combination of *-(n)un*, a case marker (i.e. *-i/ka*, *-(l)ul*, *-ey(key)*) and other focus-sensitive markers (i.e. *-man* ‘only’, *-to* ‘also’).

(3) *sala*(-n^{un}) cemsim-ul mek-ess-ta.*
Sarah-TOP lunch-ACC eat-PST-DECL
‘Sarah ate lunch.’

- Unlike written Korean, spoken Korean allows null arguments without any marking. Arguments are frequently not marked with any markers, including *-(n)un* and other case markings.

(4) *sala cemsim mek-ess-e.*
Sarah lunch eat-PST-DECL
‘Sarah ate lunch.’

- The possibility of null marking in spoken Korean allows the language to have a three-way contrast in how to mark a noun: null marking, case marking, and *-(n)un* marking. These additional types of contrast open up the possibility of having more types of readings depending on the marking.
- Importantly for this study, having an additional contrast between null marking and *-(n)un* marking results in a split between non-contrastive and contrastive use of *-(n)un*. As I will show in §3.1, this split is due to the codependent pattern between null marking and *-(n)un* marking.

(5) *sala-n^{un} cemsim-ul mek-ess-e.*
Sarah-TOP lunch-ACC eat-PST-DECL
‘SARAH ate lunch.’

Thus, in my current study I focus on the use of <i>-(n)un</i> in spoken Korean to explore this split between its two uses (i.e. non-contrastive and contrastive) via looking at the contrast between null marking and <i>-(n)un</i> marking.
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- A note on case marking: I focus on the interaction of null marking and *-(n)un* marking for this study. Case marking, especially nominative *-i/ka* which has also been argued to be pragmatic (Park 1995), provides an additional interesting layer that leads to more various readings. Though the effect of case marking is beyond the scope of this study, I would love to explore this in the future.

3. Two observations

- Main data & observations:

- (6) a. *saca-nun koki mek-e.*
 lion-TOP meat eat-DECL
 ‘Lions eat meat.’
 b. *sala-nun koki mek-ess-e.*
 Sarah-TOP meat eat-PST-DECL
 ‘SARAH ate the meat.’

- (i) The distribution of contrastive and non-contrastive *-(n)un* are co-dependent.
 (ii) The distribution of *-(n)un* is sensitive to the type of NPs that it attaches to.

3.1. Observation (i): codependency

- The distribution of contrastive and non-contrastive *-(n)un* is dependent on whether *-(n)un* is required by the NP or not.
- The marking of *-(n)un* is always non-contrastive when it is required on an NP.

- (7) a. **saca koki mek-e.*
 lion meat eat-DECL
 Intended: ‘Lions eat meat.’
 b. *saca-nun koki mek-e.*
 lion-TOP meat eat-DECL
 ‘Lions eat meat.’

- The marking of *-(n)un* is always contrastive when it is not required (i.e. null marking is allowed).

- (8) a. *sala koki mek-ess-e.*
 Sarah meat eat-PST-DECL
 ‘Sarah ate the meat.’
 b. *sala-nun koki mek-ess-e.*
 Sarah-TOP meat eat-PST-DECL
 ‘SARAH ate the meat.’

The codependent pattern between null marking and *-(n)un* marking:

- (9) * $\emptyset \Rightarrow$ non-contrastive *-(n)un*
 $\emptyset \Rightarrow$ contrastive *-(n)un*

- Thus, it is crucial to explore when an NP allows null marking to understand the effect of the additional *-(n)un* marking. This pattern requires a discussion on the interpretation of bare nouns in Korean and how it changes with the marking of *-(n)un* (further discussed in §4).

3.2. Sensitivity to NPs

- It has been previously noted that non-contrastive *-(n)un* appears in generic sentences, though there is only one paper to my knowledge that specifically discusses the occurrence of *-(n)un* in generic sentences (Lee 1996).
- Taking this as a starting point observation, I explore the distribution of *-(n)un* with respect to NP types and predicate types, mainly based on Krifka et al. (1995).

(10) Modified from Krifka et al. (1995)

		Kind-referring NP	Object-referring NP
characterizing	habitual	The lion roars when it smells food.	Simba roars when he smells food.
	lexical characterizing	The lion is a mammal.	Simba has a mane.
particular	episodic dynamic [†]	The lion disappeared from Asia	Simba roared.
	episodic statives	The lion lives in the plain.	Simba is in the cage.

[†] Also referred to as kind-level predicates

- The distribution of the two uses is sensitive to NP types that *-(n)un* attaches to, regardless of predicate types. Kind-referring NPs require obligatory *-(n)un* marking (i.e. non-contrastive *-(n)un*), whereas object-referring NPs allow null marking in which an additional *-(n)un* gives rise to contrastiveness.
- The examples below show this distribution for lexical characterizing predicates. See appendix for the other examples.

(11) a. *saca*(-nun) poyulyu-ya.*

lion-TOP mammal-DECL

‘The lion is a mammal.’

(Kind-referring NP)

b. *simba kalkwi iss-e.*

Simba mane exist-DECL

‘Simba has a mane.’

(Object referring NP)

⇒ Additional *-(n)un* on ‘Simba’ means ‘SIMBA has a mane.’

Generalization:

			kind-referring NP	object-referring NP
(12)	characterizing	habitual	obligatory <i>-(n)un</i>	allows null
		lexical characterizing	obligatory <i>-(n)un</i>	allows null
	particular	episodic dynamic	obligatory <i>-(n)un</i>	allows null
		episodic statives	obligatory <i>-(n)un</i>	allows null

⇒ Thus, this shows that the distribution of *-(n)un* does not in fact align with generic-episodic distinction, but with the NP types that *-(n)un* appears with.

4. Interpreting bare nouns

- So far, we have observed that *-(n)un* functions differently depending on what the NP it attaches to refer to (i.e. kind vs. object).
- In this section, I extend this reference to NPs to more fine-grained interpretation of NPs, focusing on the indefinite, definite and kind interpretation of bare nouns.

4.1. Bare nouns in determiner-less languages

- Some languages like English use determiners (e.g. *the/a*) to mark the (in)definiteness of a noun phrases.

- (13) a. The cat is meowing.
 ⇒ presupposes there is a unique cat. (definite)
- b. A cat is meowing.
 ⇒ presupposes there is more than one cat. (indefinite)

- Bare nouns (i.e. nouns without determiners) in determiner-less languages have been argued to be compatible with both definite and indefinite contexts.

- (14) a. *kamre meN cuuhaa hai*
 room in mouse is
 ‘There’s a mouse in the room.’
- b. *kuch bacce andar aaye. bacce bahut khush the*
 some children inside came children very happy were
 ‘Some children came in. The children were very happy.’ (Hindi; Dayal 2004)

- Bare nouns, either singular or plural, can also refer to kinds. Though Dayal (2004) notes that they have different instantiations.

- (15) a. *kutta aam janvar hai*
 dog common animal is
 ‘The dog is a common animal.’
- b. *kutte yehaaN aam haiN*
 dogs here common are
 ‘Dogs are common here.’ (Hindi; Dayal 2004)

- A side note: languages with determiners show different strategies to refer to kinds.

- (16) a. A potato contains vitamin C, amino acids, protein and thiamine.
 b. The potato is highly digestible.
 c. Potatoes are served whole or mashed as a cooked vegetable. (Krifka et al. 1995)

- Various analyses have been proposed to explain the different interpretations of bare nouns in articleless languages (Chierchia 1998, Dayal 2004).

Bare nouns in determiner-less languages, though in a rough sense, can refer to either indefinites, definites, or kinds.

- More importantly, Šimík and Demian (2020) note that factors such as word order, case, topicality seem to matter in these different interpretations of bare nouns.
- In the next section, I explore the interpretation of bare nouns in Korean regarding one of these factors, namely *-(n)un* marking.

4.2. Bare nouns in Korean

- Korean is also a determiner-less language, which has been argued that bare nouns are compatible with various readings with respect to definiteness and plurality (Kang and Park 2020).

(17) *uyca-ka pang-ey iss-ta.*
chair-NOM room-LOC exist-DECL

- a. ‘There is a chair in the room.’ [indefinite/singular]
- b. ‘There are chairs in the room.’ [indefinite/plural]
- c. ‘The chair is in the room.’ [definite/singular]
- d. ‘The chairs are in the room.’ [definite/plural]

(Kang and Park 2020)

- I first look at how bare nouns are interpreted in argument positions, mainly when used as a subject, when they do not get any marking (i.e. null-marked nouns). Then I explore how their meaning differs when they are *-(n)un*-marked (i.e. *-(n)un*-marked nouns).
- It seems to be true that null-marked nouns are compatible with contexts for both definites and indefinites, which in general follows the pattern like other determiner-less languages.

(18) *goyangi ul-goiss-e.*
cat cry-PROG-DECL

- a. ‘A cat is crying.’
- b. ‘Cats are crying.’
- c. ‘The cat is crying.’
- d. ‘The cats are crying.’

- However, bare nouns in a subject position can never refer to kinds and they always require *-(n)un* marking.

(19) *goyangi#(-nun) ul-e.*
cat-TOP cry-DECL

Intended: ‘Cats cry.’

A note on #: The sentence without *-(n)un* marking forces the bare noun to be interpreted as a definite/indefinite reading (e.g. The/A cat is crying).

It seems like *-(n)un* marking in some contexts functions as a tool for forming kinds.

- In addition, when the noun in (18) is *-(n)un*-marked, the marking not only gives rise to contrastiveness but also forces the noun to be interpreted as definite.

- (20) *goyangi-nun ul-goiss-e.*
 cat-TOP cry-PROG-DECL
 ‘The CAT is crying.’
 ‘The CATS are crying.’

- A nice contrast arises based on these observations. When bare nouns are used with predicates that are compatible with both kind and object referring NPs, a *-(n)un*-marked sentence is ambiguous between the two readings: a kind reading and a contrastive definite reading.

- (21) *goyangi-nun ul-e.*
 cat-TOP cry-DECL
 ‘Cats cry/THE CAT is crying.’

Thus, what we have seen so far regarding *-(n)un* marking is that ...

- (i) The *-(n)un* marking is obligatory for a bare noun to refer to kinds.
- (ii) The *-(n)un* marking gives rise to contrastiveness when marked on a bare noun that has a definite reading.

⇒ Though it may seem less intuitive at the first glance, the pattern of *-(n)un* suggests that there are some similarities in the interpretation of definites and kinds.

4.3. Definites and kinds

- I outline classic analyses of definites and kinds, and discuss their similarity, which in a nutshell seems to be that they both encode some meaning of maximality.

Definites

- Singular and plural definites have been argued to each encode uniqueness and maximality in their meaning, which may seem distinct.
- However, uniqueness can be understood as a type of maximality. For instance, for both singular and plural use of *the*, it presupposes the existence of a unique maximal entity. This dates back to Sharvy (1980) and Link (1983).

- (22) $\llbracket \text{the } \phi \rrbracket$ (where ϕ is of type $\langle \alpha, t \rangle$) is defined only if there is a unique maximal object x st. $\phi(x)$ is true (based on the ordering on elements of type α). The reference of *the* ϕ (when defined) is this unique maximal element. (von Stechow et al. 2014)

- Here, a maximal object is defined based on an ordering relation in the context and is an object that does not have any other object that is ordered higher.

- For singular definites, which denote an individual without parts (i.e. atomic individuals), the use of *the* presupposes the existence of a uniquely maximal atomic entity, which gives rise to a uniqueness presupposition.
- For plural definites, the ordering relations hold for the atomic entities (e.g. a, b, c) and their sums (e.g. $a + b, b + c, c + a, a + b + c$). The unique maximal object based on this ordering is $a + b + c$, which is presupposed by the use of *the*. Thus, no uniqueness presupposition arises.
- The Frege-Russell definition of ι (i.e. iota operator):

(23) ιX = the largest member of X if there is one (else, undefined)

a. the dogs = ι DOGS = the largest plurality of dogs

b. the dog = ι DOG = the only dog (if there is one) (Chierchia 1998)

Kinds

- A kind of a property denotes a ‘totality of that property’ (e.g. “the dog-kind in our world can be identified with the totality of dogs”), or in other words “a kind can be manufactured out of a property by taking the largest member of its extension (at any given world).” (Chierchia 1998)

(24) Given a property P , e.g. DOG,

A kind d , e.g. $\text{dog} = \lambda s \iota \text{DOG}_s$

Definites presuppose a maximal object while kinds denote a maximal (intensional) object.

This connection may be able to explain why in some determiner-less languages bare nouns are ambiguous between a definite and kind interpretation; this is observed with object-level predicates (Dayal 2004).

(25) *kutte bahut bhau Nkte haiN*
dogs lot bark

‘The dogs/Dogs bark a lot.’

(Hindi; Dayal 2004)

⇒ Even though definites and kinds seem very similar, intuitively their meanings are not entirely the same; definites mean something more specific and local whereas kinds mean something bigger. This leads to a question about how exactly their meanings differ. This may be explained via intensionality of kinds or some aspect of discourse restriction for definites, but further thoughts are required.

5. Proposal: maximality

- I first propose the function of $-(n)un$ based on how it derives a kind interpretation of bare nouns, then extend the analysis to explain how contrastiveness arises for definite nouns.

5.1. *-(n)un* as marking maximality

- I propose that *-(n)un* marks maximality of the interpretation of the noun it attaches to.

Deriving kinds

- When *-(n)un* attaches to a bare noun that denotes a property, it functions as the down operator, $\cap$, which maps properties to kinds.

$$(26) \quad \text{For any property } P \text{ and world/situation } s, \\ \cap P = \begin{cases} \lambda s \iota P_s, & \text{if } \lambda s \iota P_s \text{ is in } K \\ \text{undefined,} & \text{otherwise} \end{cases} \\ \text{where } P_s \text{ is the extension of } P \text{ in } s. \quad (\text{Chierchia 1998})$$

Deriving contrastiveness

- When *-(n)un* attaches to a bare noun that already denotes a maximal object (i.e. definite nouns), the additional layer of maximality gives rise to a contrastive implication; I adopt the analysis of exclusivity (Coppock and Beaver 2011), which explains how the inference of exclusivity is conventionalized and calculated via maximalization.
- To elaborate, this maximalization analysis can explain the conventional meaning of *-(n)un* that has been proposed by Park (2023).
- In Park (2023) I have argued that the core conventional meaning of contrastive *-(n)un* is to encode “inadequacy”, which implies incompleteness with respect to the information that is being shared by the speaker at the current state of discourse. This is elaborated below.

$$(27) \quad \text{For a } -(n)un\text{-marked sentence of the form “}x-(n)un P\text{”, } -(n)un \text{ conventionally encodes} \\ \begin{aligned} &a. \text{ that } x \text{ is the maximal entity that holds true for } P \text{ among the set of information} \\ &\quad \text{which will be shared by the speaker.} \\ &b. \text{ that there are alternatives } y \text{ to } x \text{ such that the information that “}y \neg P\text{” is possibly} \\ &\quad \text{relevant to the context but is not being provided by the speaker.} \end{aligned}$$

- This conventional meaning gives rise to three contextually-determined sub-implications (i.e. uncertainty, exhaustivity, unwillingness), which have been observed in the literature.

5.2. But why?

- Now the important question is why bare nouns in Korean are interpreted differently, and how we can explain why the maximality of *-(n)un* leads to two very different interpretations (i.e. kind and contrastive).
- One (unsatisfactory) explanation is that bare nouns in Korean can only denote indefinites and definites, and never kinds.

A typology of kind reference

- The interpretations of bare nouns in Korean do not seem to fit in any of the categories of languages (adapted from Chierchia 1998).

- The closest possible category is a Chinese-type language since Korean allows bare nouns relatively freely. Additionally, Korean does not require a plural marking and uses a classifier system as well.

(28) Chinese-type languages

- i. Use of bare nouns
- ii. No obligatory plural marking
- iii. Classifier system

c.f. Romance-type languages are observed to disallow bare nominal arguments, and English-type languages allow bare nominal arguments for mass nouns and kinds, though only for plurals.

- However, this categorization does not explain why a bare noun in Korean cannot by itself denote a kind.

I wish I had a simple definition for *-(n)un* and the clearer picture on bare noun interpretations in Korean that neatly explains the interaction of *-(n)un* and bare nouns. However, based on what I have so far, I can at best propose two possible explanations.

An elsewhere condition (less satisfactory explanation)

- We can assume a conditioned analysis:
 1. If a bare noun denotes a definite meaning, the maximality of *-(n)un* gives rise to a contrastive implication like exclusivity.
 2. Elsewhere, the maximality of *-(n)un* results in a semantics that forms a kind.

Maximality as redundancy (slightly better explanation)

- We can assume that *-(n)un* has a main function of forming a kind.
- The interaction of *-(n)un* with definites can be explained as a special case showing a discourse effect. For instance, in some cases, the infelicity of a sentence gets resolved in a specific context where the interlocutors reason why a speaker may have said that sentence.

(29) a. # Mary loves Jane, therefore somebody loves Jane.

b. A: No one loves Jane.

B: Mary loves Jane, therefore somebody loves Jane.

(Beavers p.c.)

- As for *-(n)un* which seemingly functions as a redundant marking of maximality when marked on definites, we can say that its function diverges from its main use due to these discourse effects, which leads to a contrastive implication.

6. Conclusion

- I showed that the *-(n)un* marking exhibits two seemingly distinct functions: (i) it marks a noun to be interpreted as kinds and (ii) it gives rise to a contrastive implication when the noun specifically denotes a definite reading.
- I proposed the maximality function for *-(n)un* which is sensitive to the interpretation of the noun it attaches to. This works as a link that connects the two distinct functions of *-(n)un*.
- I gave two possible explanations for the two interpretations of the maximality of *-(n)un*, though they require further thoughts.
- Further issues that I am currently working on:
 1. The distribution of *-(n)un* in more fine-grained predicate types (e.g. dispositionals and habituels).
 2. The interpretation of bare nouns in object positions and its interaction with *-(n)un* marking.

Appendix: Further examples for §3.2

(30) Characterizing/habitual

- a. *saca*(-nun) umsik nemsey mat-emyun ul-e.*
 lion-TOP food smell(n.) smell(v.)-if cry-DECL
 ‘The lion roars when it smells food.’ (Kind-referring NP)
- b. *simba umsik nemsey mat-emyun ul-e.*
 Simba food smell(n.) smell(v.)-if cry-DECL
 ‘Simba roars when it smells food.’ (Object-referring NP)

(31) Particular/episodic dynamic

- a. *saca*(-nun) asia-eyse salacy-ess-e.*
 lion-TOP Asia-LOC disappear-PST-DECL
 ‘The lion disappeared from Asia.’ (Kind-referring NP)
- b. *simba ul-ess-e.*
 Simba cry-PST-DECL
 ‘Simba roared.’ (Object-referring NP)

(32) Particular/episodic statives

- a. *saca*(-nun) pyungwon-eyse sal-a.*
 lion-TOP plain-LOC live-DECL
 ‘The lion lives in the plain.’ (Kind-referring NP)
- b. *simba uri-ey iss-e.*
 Simba cage-LOC exist-DECL
 ‘Simba is in the cage.’ (Object-referring NP)

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